

ANALYSTLAB AFRICA — HEALTHCONNECT EXPERIENCE LAB

Week 8: HealthConnect Final Integration, Presentation & Project Showcase

Data Analytics Track — Final Package

Prepared by: Omolade

Part 0: Week 7 → Week 8 Transition

Requirement	Response
Main Week 7 output	The Week 7 Testing & Refinement Report — validated Week 6 KPIs, tested the composite risk score across subgroups, and produced a corrected Data Science handoff file (v2).
Most important testing result	The original 0–5 risk score's no-show gradient broke down in 4 of 13 subgroups; the simplified 3-tier band (Low/Medium/High) passed monotonicity in all 13.
Major issue discovered	The handoff CSV's reminder_channel column had 1,366 missing values. A first fix (filling with the text "None") silently failed, because pandas reads "None" back as missing.
What was refined/improved	Collapsed the risk score to 3 tiers; relabeled the missing reminder_channel values as "No Reminder Sent"; published handoff file v2.
Which component is now ready	The 3-tier risk band and the v2 handoff CSV — both retested and validated.
What still needs to be integrated	Final confirmation from Data Science that v2 resolves their loading issue; folding the validated risk band into the final dashboard/report for presentation.
Which track(s) involved in final integration	Data Science (primary); Project Management (coordination for the final presentation).
What to demonstrate in final presentation	How raw appointment data became validated no-show/risk findings, and how those findings turned into a clinic-ready 3-tier risk indicator that Data Science can safely build on.

Part 1: Final Integration Readiness

Requirement	Your Response
What is my final component?	The HealthConnect 3-tier Patient No-Show Risk Band (Low/Medium/High), plus the finalized analytics dashboard/report.
What problem does it address?	Missed appointments and inefficient use of slots — specifically, identifying which patients are most likely to no-show so the clinic can intervene proactively.
What is its current status?	Conditionally Ready for final integration — retested across 13 patient subgroups with monotonic separation confirmed in every subgroup, and independently corroborated by Data Science's own model feature-importance ranking (a completely different method arriving at the same top predictors). Full readiness depends on one small, well-scoped remaining step: Data Science's follow-up test of a proposed interaction feature.
What did I improve during Week 7?	Collapsed the fragile 0–5 score into a statistically robust 3-tier band; fixed a real missing-value bug in the Data Science handoff file; directly

Requirement	Your Response
	cross-validated key findings with a Data Science track partner and surfaced a new compound-risk segment (2+ prior no-shows AND 25.8km+ distance = 80.0% no-show rate) not yet captured as its own feature in their model.
Which other track(s) does it depend on?	None upstream — Data Analytics works directly from the raw HealthConnect Appointment Data.csv.
Which track(s) depend on my output?	Data Science (uses the risk band/handoff file and the feature-relevance findings); indirectly, Generative AI and Project Management (recommendations inform the final HealthConnect narrative).
What output will I provide?	HealthConnect_RiskScore_Handoff_for_DataScience_v2.csv, the Final Analytics & Decision Support Report, and the finalized dashboard.
What output do I need from another track?	Confirmation from Data Science that v2 of the handoff file loads and performs as expected in their model, and the result of their follow-up test on the proposed prior-no-shows × distance interaction feature.
What evidence demonstrates readiness?	Independent recomputation matching reported figures exactly; 13/13 subgroup monotonicity pass; 0 missing values confirmed via value counts on reload; independent corroboration from Data Science's feature-importance ranking (booking_lead_days and distance_to_clinic_km rank #1 and #2 on their side, matching my chi-square rankings exactly).
What limitations remain?	Small-sample subgroups (n=108 "Prefer not to say"; 18–24 and 65+ age bands) should be read cautiously. distance_to_clinic_km (1.8% missing) and waiting_time_minutes (1.2% missing) remain genuinely missing — flagged, not resolved. The extreme compound-risk segment (n=20) is too small to test further without more data.

Part 2: HealthConnect Final Analytics & Decision Support Report

Executive Summary

Across the HealthConnect Experience Lab, the Data Analytics track's core finding is that patient no-shows are not random — they cluster tightly with the composite risk indicator built and validated during Weeks 6–7. The clinic's overall no-show rate of 48.46% (independently recomputed in Week 7) is too high to treat as a single, undifferentiated problem. The final validated 3-tier risk band proves the gap is real and actionable: patients in the Low tier no-show at 22.8%, while High-tier patients no-show at 63.7%. This gives HealthConnect a simple, statistically defensible operational tool — direct proactive outreach toward High-tier patients first.

Final Validated KPIs

KPI 1 — Overall No-Show Rate

- 48.46% (2,423 of 5,000 appointments) — independently recomputed in Week 7; matches Week 5/6 reported 48.5%. The central baseline figure for the whole project.

KPI 2 — No-Show Rate by Reminder Channel

- None: 51.4% · WhatsApp: 49.8% · Email: 48.4% · SMS: 45.75%
- Checked for confounding in Week 6 — channel usage is nearly identical across all booking lead-time bands, so the SMS effect is real and not explained by lead time.

KPI 3 — No-Show Rate by Booking Lead Time

- Same-day: 24.7% · 1–7 days: 28.3% · 8+ days: 51.5%

- Statistically validated in Week 6 — chi-square 130.7, $p < 0.001$ (highly significant, not due to chance).

KPI 4 — Average Distance & Waiting Time by Outcome

- Distance to clinic: Attended 9.67km · No-Show 10.53km · Cancelled 10.11km
- Waiting time: Attended 24.3min · No-Show 24.2min · Cancelled 23.2min
- Confirmed negligible effect — neither is a meaningful driver of no-shows in this dataset.

KPI 5 — Repeat No-Show Rate

- Has previous no-shows: 55.4% (1,152 of 2,079) · No previous no-shows: 43.5% (1,271 of 2,921)
- Statistically validated in Week 6 — chi-square 69.9, $p < 0.001$.

KPI 6 — Composite Risk Score Gradient (original 0–5 scale, Week 6)

- Score 0: 11.8% (n=34) · 1: 24.1% (n=282) · 2: 44.3% (n=2,107) · 3: 52.6% (n=2,013) · 4: 62.9% (n=542) · 5: 81.8% (n=22)

KPI 7 — Final 3-Tier Risk Band (Week 7 refinement — operational metric)

- Low (0–1): 22.8% · Medium (2–3): 48.3% · High (4–5): 63.7%
- Validated as monotonically increasing across all 13 tested patient subgroups (age group, gender, appointment type) — the recommended version going forward.

Reminder Channel Distribution (n=5,000)

- SMS 2,000 · No Reminder Sent 1,366 · WhatsApp 1,101 · Email 533

KPI 8 — Compound Risk Segment (Week 7 cross-track validation, new finding)

- Patients with 2+ prior no-shows AND 25.8km+ distance to clinic: 80.0% no-show rate — the single highest-risk segment identified in the entire analysis, though based on a small sample (n=20).

Validated No-Show & Attendance Patterns

- No-show risk increases monotonically with the composite risk score in every one of 13 patient subgroups tested (age group, gender, appointment type) — the 3-tier pattern holds for essentially every kind of patient in the dataset, not just in aggregate.
- Booking lead time is the single strongest standalone factor: appointments booked 8+ days ahead are more than twice as likely to be missed as same-day bookings (51.5% vs 24.7%), and the relationship is statistically significant ($p < 0.001$).
- Prior no-show history is the second-strongest standalone factor: patients with any no-show history are nearly 12 points more likely to miss again (55.4% vs 43.5%, $p < 0.001$).
- These two factors compound rather than simply add up: patients booking 8+ days ahead with a prior no-show history have a 58.3% no-show rate (n=1,811), versus just 18.4% for same-day bookings with no prior history (n=49) — a more than 3x gap between the best- and worst-case combination.
- SMS remains the dominant and most reliable reminder channel, and this holds independent of booking lead time — a Week 5/6 finding unaffected by Week 7 testing.
- Distance to clinic and waiting time have negligible effect on attendance — access/logistics are not the primary driver of no-shows in this dataset.
- 1,366 appointments (about 27% of the dataset) had no reminder sent at all — a large, addressable gap and a natural first lever for the clinic, separate from risk-based targeting.

Final Business Insights

- 1. Segmentation works:** the validated 3-tier risk band cuts the no-show rate from 63.7% down to 22.8% just by grouping on already-known patient information.

2. Risk factors compound: a patient who books far ahead AND has a no-show history is over 3x more likely to miss their appointment than one with neither factor (58.3% vs 18.4%) — these aren't two separate weak signals, they multiply.

3. High-tier patients are the priority: this holds true across every age group, gender, and appointment type tested — it's now backed by subgroup-level evidence, not just an aggregate trend.

4. The reminder gap is low-cost, high-leverage: roughly 1 in 4 appointments gets no reminder — closing this gap doesn't require the risk model at all.

5. Access isn't the problem: distance and waiting time show almost no relationship with no-shows — HealthConnect's resources are better spent on scheduling and reminder behavior than on clinic logistics.

6. Data quality discipline pays off cross-track: the reminder_channel fix (relabeling instead of blank-filling) caught an issue that would have silently broken Data Science's model input.

7. Simplicity beat precision: the 3-tier band, despite less granularity than the original 0–5 score, is the more trustworthy and communicable tool — a useful lesson for how HealthConnect should operationalize future scoring work.

8. Independent corroboration strengthens the whole analysis: Data Science's own model ranks booking_lead_days and distance_to_clinic_km as its #1 and #2 most important features — matching my chi-square rankings exactly — and previous_no_shows sits in their top 10. Two unrelated methods reaching the same conclusion is stronger evidence than either alone.

9. A gap worth closing: the compound segment of 2+ prior no-shows AND 25.8km+ distance no-shows at 80.0%, HealthConnect's single highest-risk group — yet Data Science's model has no feature that captures this interaction, despite having both variables separately.

Final Recommendations

1. Adopt the 3-tier risk band (Low/Medium/High) as HealthConnect's standard operational tool for flagging at-risk appointments, not the original 0–5 score.

2. Prioritize High-tier patients for proactive outreach — calls, confirmed bookings, flexible rescheduling — given the validated 63.7% no-show rate in that group, and especially those who also booked 8+ days out with a prior no-show history (58.3% no-show rate).

3. Close the reminder gap: extend SMS reminders to the 1,366 appointments (about 27%) currently receiving none — SMS is the validated best-performing channel.

4. Deprioritize clinic-access investments: reducing waiting time or distance is not an effective no-show lever — redirect that budget toward reminder intensity and scheduling interventions instead.

5. Monitor, don't ignore, small subgroups: "Prefer not to say" gender and the 18–24 / 65+ age bands have small extreme-score sample sizes; keep tracking as more data accumulates.

6. Standardize the v2 handoff format (explicit labels like "No Reminder Sent" instead of blank/"None") as the template for future cross-track data exchange.

7. Add the interaction feature: recommend Data Science test an explicit prior-no-shows × distance feature — this would let the model directly capture its highest-risk segment (80.0% no-show rate) instead of relying on the two variables separately.

Cross-Track Contribution

This track's most significant final contribution is the validated risk band and the corrected v2 handoff file delivered to Data Science — closing a real data-integrity gap before it could affect model training. Full detail is documented as the mandatory HC-POD integration record in Part 3.

Limitations

- distance_to_clinic_km (1.8% missing) and waiting_time_minutes (1.2% missing) remain genuinely missing — flagged for Data Science's modelling decision, not resolved this cycle.
- Small-sample subgroups should still be read with some caution, even under the 3-tier band.
- Findings are based on the fictional HealthConnect_Appointment_Data.csv dataset only and would need real clinic data before any operational deployment.

Part 3: HC-POD Final Integration Record — Data Analytics → Data Science

Integration Activity 1 — Risk Score Handoff File

Item	Response
1. Track collaborated with	Data Science
2. Dependency	Data Science's predictive model consumes the composite risk indicator / handoff file as an input or validation reference.
3. Output received	Handoff file was tested as-loaded rather than requiring input back from Data Science at this stage.
4. Output provided	HealthConnect_RiskScore_Handoff_for_DataScience_v2.csv (0 missing values in reminder_channel; distance/waiting-time gaps explicitly flagged), plus the validated 3-tier risk band methodology.
5. Final integration activity	Re-tested the v2 handoff file as if loaded by Data Science's own workflow — full reload plus a missing-value check — confirming it is safe for direct pandas use.
6. What changed	reminder_channel's 1,366 missing values were relabeled from a silently-broken "None" fix to an explicit "No Reminder Sent" label.
7. How the change improved the solution	Removed a hidden data-quality trap that would have broken Data Science's model training without warning — caught only by re-testing the actual load behavior, not by inspecting file structure.
8. Evidence	Before/after value counts (SMS 2,000 / No Reminder Sent 1,366 / WhatsApp 1,101 / Email 533); 0 missing values in reminder_channel confirmed on reload.
9. Contribution to final presentation	A concrete example of "actual use of another track's output" — a real bug caught and fixed through cross-track testing, not just communication.

Integration Activity 2 — Cross-Track Feature Validation

Item	Response
1. Track collaborated with	Data Science (direct exchange with a track partner, via cold outreach after initial cohort messages went unanswered).
2. Dependency	Two-way: Data Science's model relies on shared candidate features that Data Analytics identified; Data Analytics' confidence in its own findings is strengthened by independent confirmation from Data Science's model.
3. Output received	Data Science's Week 6 Integration & Progress Questionnaire (baseline model details, feature list, evaluation metrics, limitations), plus direct answers to four specific questions on feature usage, leakage handling, and Cancelled-class treatment.
4. Output provided	Validated KPI findings, the waiting_time_minutes leakage caution, the age × appointment-type revision, and a reciprocal statistical check on Data Science's two flagged weak segments (Specialist Consultation,

Item	Response
	Sunday appointments).
5. Final integration activity	Two-way convergent validation: cross-checked feature-relevance rankings against Data Science's actual model decisions, and reciprocally tested Data Science's two weak segments against the Data Analytics dataset.
6. What changed	Full convergence confirmed on shared top features (see evidence). A new compound-risk finding surfaced a gap in the Data Science model — no interaction feature for 2+ prior no-shows AND 25.8km+ distance (80.0% no-show rate). Data Science's two weak segments were shown not to reflect a real underlying pattern ($p=0.50$, $p=0.24$).
7. How the change improved the solution	Confidence in the key findings increased because they're now confirmed by two independent methods (chi-square testing and Data Science's model feature importance) rather than one. Data Science gained a concrete, testable recommendation (the interaction feature) and a data-backed reason to deprioritise two segments they were investigating as weaknesses.
8. Evidence	CHISQ.TEST calculations (Excel) for all 4 tested findings plus 2 reciprocal segment checks; a Statistical Validation table on the Power BI dashboard; before/after dashboard screenshots; Data Science's Week 6 Integration & Progress Questionnaire; the documented message exchange.
9. Contribution to final presentation	Demonstrates genuine two-way cross-track validation — not just a data handoff, but mutual confirmation and a concrete follow-up recommendation, meeting the HC-POD requirement that collaboration involve actual exchange and use of outputs, not communication alone.